indeterminism are not forced on us by experimental facts, but by deliberate theoretical choice?" But not long after Bohm returned to pilot-wave theory, Bell took up the banner of one of the newer ideas being developed at the time: *spontaneous-collapse theory.*

Rather than interpreting the existing mathematics of quantum physics, as Bohm and Everett had done, spontaneous-collapse theory actually modifies the equations of quantum physics to solve the measurement problem. It does so in a subtle way—as it would have to, since quantum physics is spectacularly successful in predicting the outcomes of experiments. But spontaneous-collapse theory manages to leave most of the predictions of standard quantum physics intact, while altering them enough to solve the measurement problem.

In spontaneous-collapse theory, the quantum wave function is real, but it doesn't obey the Schrödinger equation perfectly. Instead, sometimes the wave function collapses. But this collapse has nothing to do with observation or measurement—the collapse happens entirely at random, for no reason at all, whether or not anyone is looking. Think of the wave function as playing a collapse slot machine (Figure 10.3a): every time the wave function hits the jackpot, it collapses. It pulls the handle millions of times a second, but the collapse jackpot only comes up once every 10 *million billion billion* times or so—a one with 25 zeros after it—so it takes hundreds of billions of years for the wave function to collapse. This means that subatomic particles can almost always go down two paths at once, just like our nanometer Hamlet in the Introduction—but every once in a very great while, they're forced to a single path. (Just how great a while is a matter to be resolved by experiment, but it must be at least tens of thousands of years, otherwise the theory would contradict existing experiments.)

Yet this still leaves us with the question from the Introduction: if subatomic particles can behave so strangely and we and the objects in our everyday lives are composed of such particles, why don't we see such strange behavior on a regular basis? According to spontaneous-collapse theory, the answer lies in two key facts: entanglement and the vast number of particles that comprise the objects of our everyday experience. Though a single-particle wave function might not collapse on average until a billion years have passed, the solid objects of our everyday lives, like this book,